

WEEK-3 [4-QUEENS]

Code:

```
import random

N = 4

def cost(state):
    conflicts = 0
    for i in range(N):
        for j in range(i + 1, N):
            if state[i] == state[j]:
                conflicts += 1
            if abs(state[i] - state[j]) == abs(i - j):
                conflicts += 1
    return conflicts

def get_neighbors(state):
    neighbors = []
    for col in range(N):
        for row in range(N):
            if row != state[col]:
                new_state = state.copy()
                new_state[col] = row
                neighbors.append(new_state)
    return neighbors

def hill_climbing():
    current = [random.randint(0, N - 1) for _ in range(N)]
    while True:
        current_cost = cost(current)
        print("State:", current, "Cost:", current_cost)
        if current_cost == 0:
            return current
        neighbors = get_neighbors(current)
        best = min(neighbors, key=cost)
        if cost(best) >= current_cost:
            return None
        current = best

while True:
    solution = hill_climbing()
    if solution:
```

```
print("\nSolution:", solution)  
break
```

Output :

```
State: [1, 1, 0, 1] Cost: 5  
State: [1, 1, 0, 2] Cost: 2  
State: [1, 3, 0, 2] Cost: 0  
  
Solution: [1, 3, 0, 2]
```